Reliable Multi-Scale Sensor Fusion Framework for Agricultural Robots

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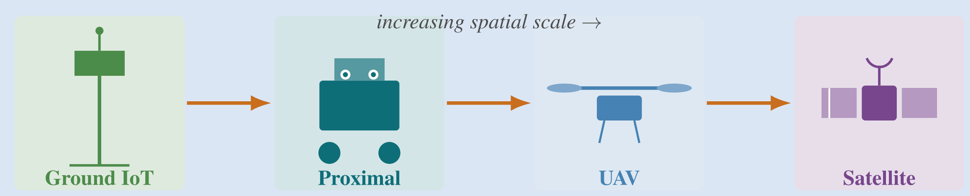
Problem

Robots need crop & field state across many scales — one plant to a whole region.



- No single sensor sees all scales.
- Demands principled **multi-scale fusion**.

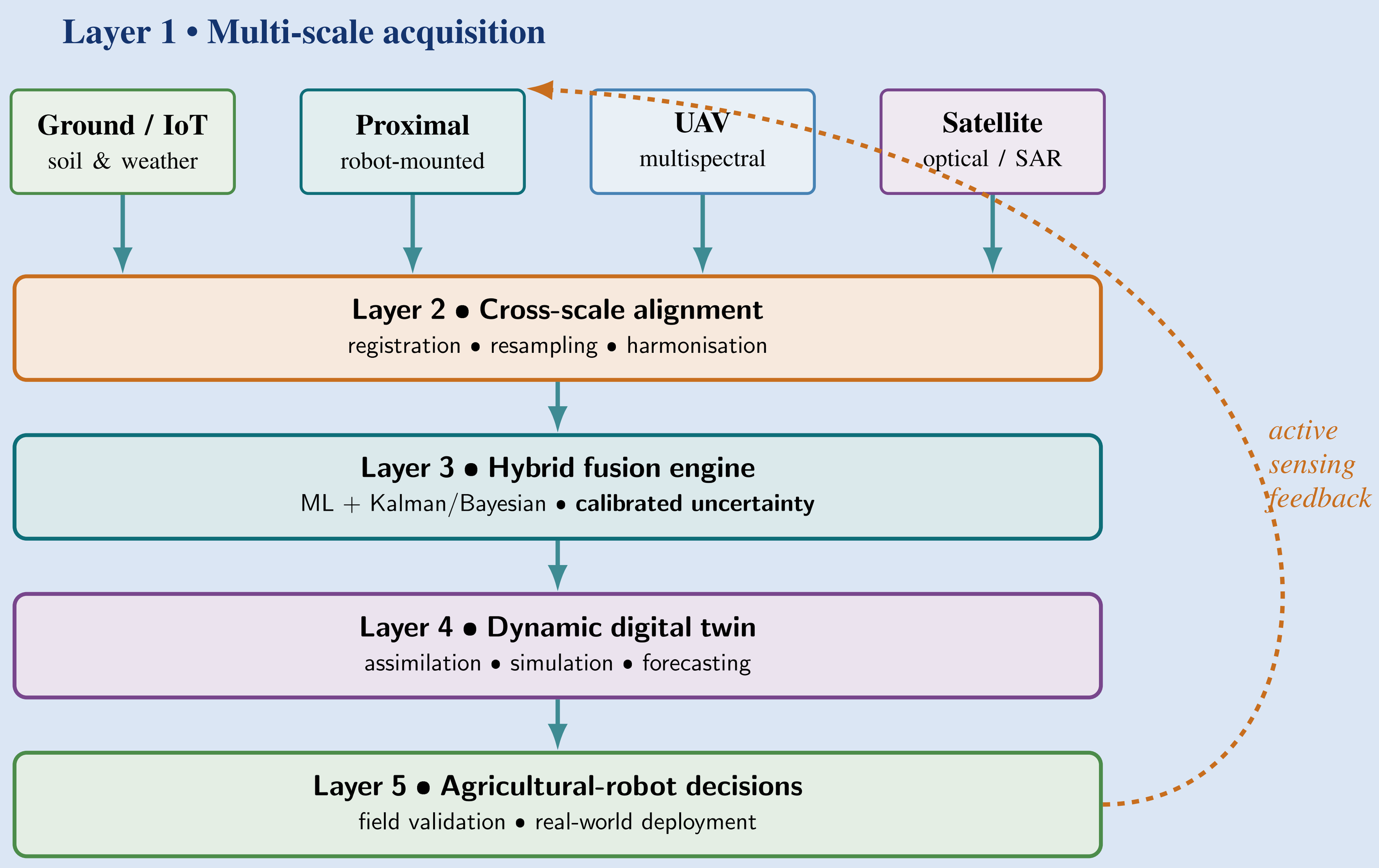
Goal

A **reliable multi-scale fusion framework**:
- Fuse **ground IoT** · **proximal** · **UAV** · **satellite**.
- **Calibrated uncertainty** + dynamic **digital twin**.
- Validated **in the field**.

State of the Art & Gap

Fusion is mature (Bayesian/Kalman → deep learning) [1]; agri remote-sensing fusion [2] & digital twins [3] are growing.
- Mostly **single-scale**, **offline**.
- Reliability under **degraded data** ignored.
- Perception→twin→robot loop unexplored.

Proposed Framework



Objectives

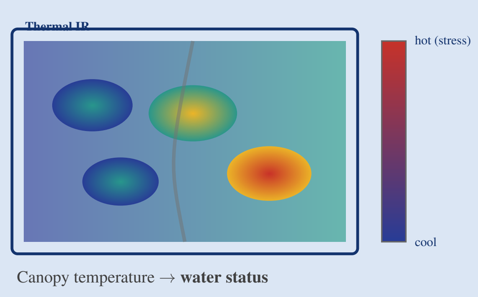
O1 Systematic review.
O2 IoT pipeline & alignment.
O3 Baseline fusion prototypes.
O4 Uncertainty-aware engine.
O5 Couple to digital twin.
O6 Field & robot validation.

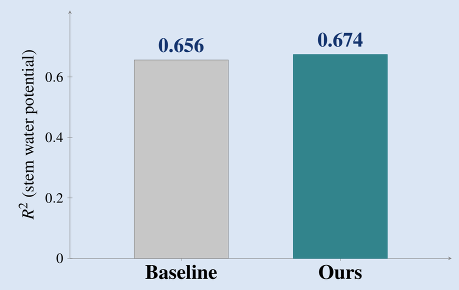
Methodology

- **Align:** registration & harmonisation.
- **Fuse:** ML + Kalman/Bayesian.
- **Reliability:** calibrated, robust to dropout.
- **Twin:** simulate & forecast.
- **Validate:** accuracy & robustness.

Pilot study area

Preliminary Results



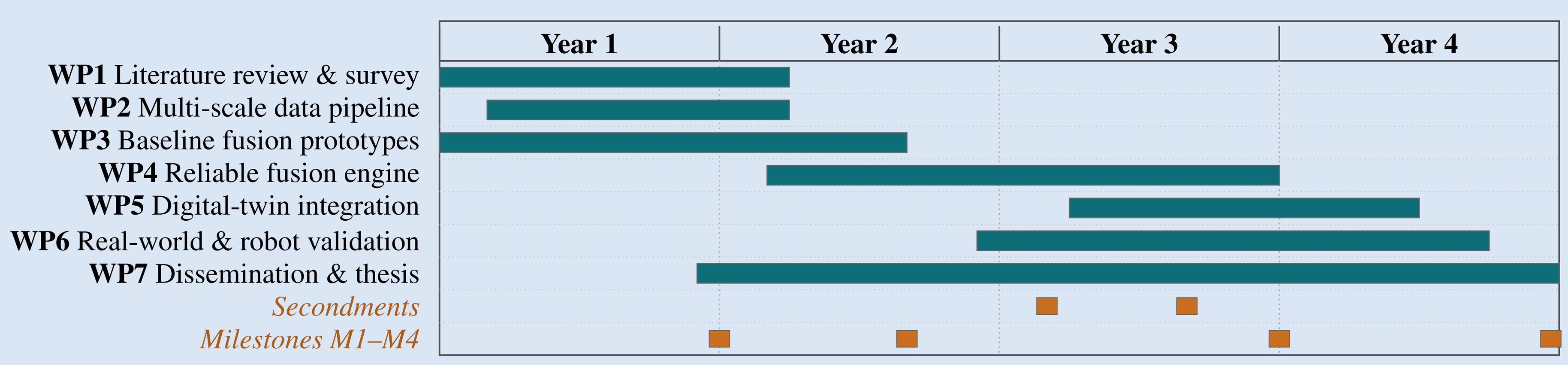


Ground-scale **thermal pilot**: >**87%** wins · skewness ↓**26–55%** · variety-agnostic.

Expected Contributions

- Uncertainty-aware fusion framework.
- IoT pipeline & aligned dataset.
- Hybrid ML + probabilistic fusion.
- Digital twin + active sensing.
- Real-world robot validation.

Work Plan (48 months)



References

[1] Khaleghi *et al.*, *Information Fusion*, 2013.
[2] Ghamisi *et al.*, *IEEE GRSM*, 2019.
[3] Peladarinos *et al.*, *Sensors*, 2023.
[4] Pires *et al.*, *Comput. Electron. Agric.*, 2025.

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